RTTI - Speed limits

Introduction to semantic modelling

A **concept scheme** (in SKOS: *ConceptScheme*) is a structured collection of concepts that carry a fixed, shared meaning within a particular domain. The goal is for everyone working with the model — people and systems alike — to use the same words for the same things. That sounds obvious, but in practice language causes a surprising amount of confusion: what one organisation calls a *measurement point*, another calls a *sensor*, a *location*, or an *observation station*.

A concept scheme solves this by recording, for each concept:

- a **preferred label** (prefLabel) it is known by,
- a **definition** that unambiguously describes what the concept does and does not cover,
- and optionally a **scope note** or **example** to clarify borderline cases.

Concepts rarely stand on their own. They are related **hierarchically** (a vehicle is broader than a truck) or **associatively** (a road user relates to a road segment). These relationships make it possible for queries and analyses to navigate between general and specific concepts, and to make connections between concepts explicit. Within SKOS we use `broader`, `narrower`, and `related` for that.

When a concept also exists elsewhere — in a national or European standard, for instance — we record that as a **mapping relation**. This prevents reinventing the wheel: we link to the source so interoperability with external systems remains possible.

This document describes the concept scheme as it is currently recorded. The following sections list the metadata, every concept, the relationships between them, and — where the model is connected to a data source — how it is linked to that source data.

RTTI - Speed limits

Description

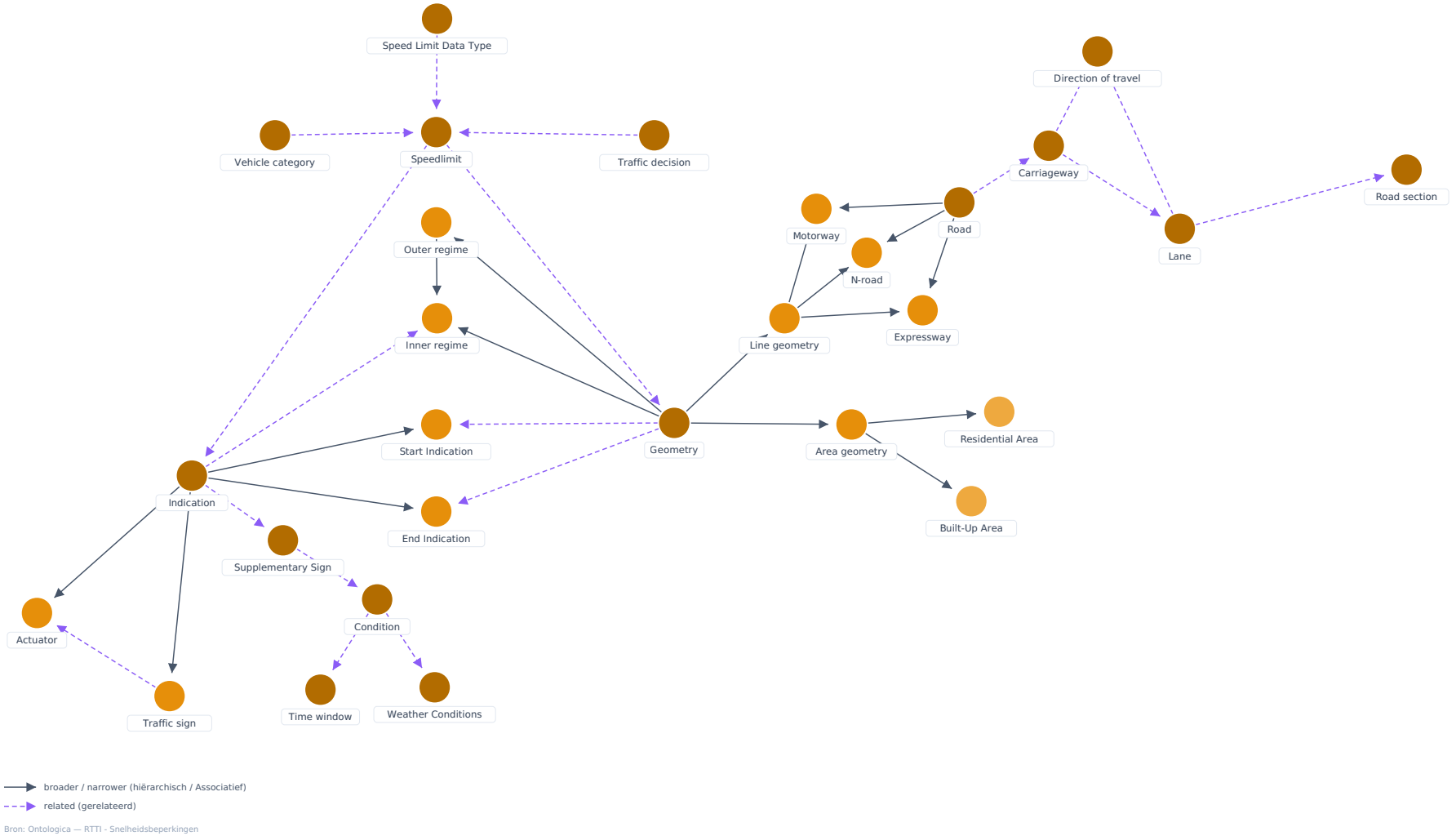
This vocabulary describes the information needed to record how fast traffic may travel on a road. The core is the maximum speed, which may follow from the statutory default (for example 50 km/h within built-up areas, 80 km/h on other roads outside them, 100 km/h on expressways and 130 km/h on motorways), from a traffic sign, or from a digital sign that adjusts the speed on the spot. For each speed limit, the vocabulary also records: where it applies (which stretch of road, which carriageway or lane, or within which area/zone), how road users see it (through ordinary roadside signs, zone signs, supplementary signs or electronic matrix signs), when it applies (always, within a certain time window, or depending on conditions such as the weather), which vehicles it applies to (some vehicle categories have their own maximum speed), what exceptions there are, and the official decision the rule is based on. The model also accounts for zones nested within one another: entering a 30 km/h zone inside a built-up area makes 30 apply temporarily, and on leaving it the speed of the enclosing zone applies again. By describing all of this in a consistent way, different parties — such as road authorities, navigation systems and travel information services — can reliably exchange and use the information, including for in-car speed information. The vocabulary is part of the European arrangements for travel and traffic information (the ITS Directive 2010/40/EU and the RTTI Regulation (EU) 2022/670) and connects to existing Dutch rules and sources for roads and traffic signs (including the RVV 1990, NDW and the National Road Database).

Metadata

Field	Value
URI	https://ndwnu.github.io/its-gegevenscatalogus/id/9a2ca40b-96c1-4526-a5c5-fda1a468fc5b
Version	0.7.0
Publication status	Draft
Management status	Active
Organization	Nationaal Toegangspunt Mobiliteitsdata
Created on	2026-02-16
Last modified on	2026-10-06
Last modified by	arthur.keesen@ndw.nu

Model overview

Visual overview of the concepts and their relations. Hierarchical relations are drawn with solid arrows; associative relations are drawn with dashed arrows.



Concepts

Outer regime

Concept details

Field	Value
Definition	A demarcated area within which a specific speed limit is in force. Zones can be nested: a zone can lie within an enclosing zone, so that multiple speed limits apply simultaneously in a cascading structure. On leaving an inner zone, the speed limit applying there lapses and the applicable speed falls back to that of the enclosing zone.
Example	A road user enters a built-up area (50 km/h). Subsequently, they enter a residential area (15 km/h). The active speed restriction is now 15 km/h. Upon leaving the residential area, 50 km/h applies again, because the built-up area as enclosing zone is still in effect.

Road section

Concept details

Field	Value
Definition	A section of road between two consecutive junctions or intersections, or – if no side road is present – between two points to which a traffic measure applies. In the National Road Database (NWB) a road section is the smallest unit for which road attributes are registered.

Residential Area

Concept details

Field	Value
Definition	A part of the public road designated as a residential area (erf), bounded by sign G5 (start of residential area) and sign G6 (end of residential area). Within a residential area a maximum speed of 15 km/h applies.

Speedlimit

Concept details

Field	Value
Definition	The permitted speed on a road or road section, expressed in kilometres per hour (km/h). In the Netherlands the maximum speed is determined by: (1) the statutory default limits (2) by means of traffic signs (3) dynamically set limits via digital signs.

Expressway

Concept details

Field	Value
Definition	Road indicated by sign G3. Outside built-up areas a statutory maximum speed of 100 km/h applies to motor vehicles on expressways unless otherwise indicated. Only motor vehicles that are able and permitted to drive at least 50 km/h may use expressways.

Motorway

Concept details

Field	Value
Definition	Road indicated by sign G1. Outside built-up areas a statutory maximum speed of 130 km/h applies to motor vehicles on motorways unless otherwise indicated. Only motor vehicles that are able and permitted to drive at least 60 km/h may use motorways.

Carriageway

Concept details

Field	Value
Definition	Any part of the road intended for moving traffic, with the exception of cycle paths and cycle/moped paths.

Lane

Concept details

Field	Value
Definition	Part of the carriageway, marked by continuous or broken lines, of such a width that drivers of motor vehicles on more than two wheels can make use of it. When a traffic sign is applied to specific lanes, a speed limit can be restricted to one or more lanes, for example at rush-hour lanes or matrix signs.

Vehicle category

Concept details

Field	Value
Definition	Classification of a vehicle based on the statutory categorisation in the RVV 1990 and the Vehicle Regulations. Relevant for speed limits because the law establishes special maximum speeds per vehicle category.

Weather Conditions

Concept details

Field	Value
Definition	Meteorological conditions (such as rain, fog, snow, black ice or storm) that give rise to an adjusted speed limit. In the Netherlands weather-dependent speed limits are mainly set dynamically via matrix signalling on motorways (RWS – Praktijkrichtlijn Dynamisch Verkeersmanagement) or indicated statically via supplementary signs (e.g. 'when the road surface is wet').

Start Indication

Concept details

Field	Value
Definition	The traffic sign indicating the point at which the validity of a speed limit begins.

End Indication

Concept details

Field	Value
Definition	The traffic sign indicating the point at which the validity of a speed limit ends.

Indication

Concept details

Field	Value
Definition	The tin or digital sign by which a speed limit is communicated to road users.

Actuator

Concept details

Field	Value
Definition	An electronic signalling sign on which traffic signs can be displayed dynamically. A digital sign can indicate a lower speed than the static ('tin') sign; in that case the lower speed of the digital sign applies. Only when the digital sign shows an end indication or is blank does the speed limit of the static sign revive. Digital signs are used in particular on motorways via matrix signalling (RWS).

Condition

Concept details

Field	Value
Definition	A condition that modifies the standard speed limit on a road section. In Dutch practice such conditions are communicated via supplementary signs and may relate to vehicle type, time window, weather conditions or other circumstances.

Supplementary Sign

Concept details

Field	Value
Definition	A sign mounted below a traffic sign to provide further specification of the effect of that traffic sign. Legislation determines which types of supplementary signs are permitted, including: an indication of the type of road user to which the sign applies, a time indication, or a distance indication.

Built-Up Area

Concept details

Field	Value
Definition	The area established as a built-up area by the municipal council. The built-up area is indicated by signs H1 (start of built-up area) and H2 (end of built-up area). A speed limit of 50 km/h applies unless otherwise indicated by traffic signs.

Inner regime

Concept details

Field	Value
Definition	A zone nested within an enclosing zone, whose speed limit takes precedence over that of the enclosing zone for as long as the road user is within the inner zone.
Scope	The inner zone concept describes the cascading effect of nested zones. When entering an inner zone, the speed restriction of the enclosing zone is temporarily suspended. When exiting, the speed restriction of the enclosing zone becomes active again. This mechanism works recursively: multiple levels of nesting are possible.

Geometry

Concept details

Field	Value
Definition	A spatially demarcated area within which a uniform traffic regime applies, bounded by traffic signs indicating the start and end of the regime. Within the context of speed limits, a closed zone is an area whose boundaries are marked by zone signs (e.g. sign A01-30-ZB / A01-30-ZE for a 30 km/h zone, or sign G5/G6 for a residential area).

Speed Limit Data Type

Concept details

Field	Value
Definition	Data type within the RTTI context describing the set of concepts required to unambiguously record and exchange a speed limit on the Dutch road network. Encompasses the value of the speed limit, the location, the validity, the indication and any exceptions.

Traffic sign

Concept details

Field	Value
Definition	The physical object — a panel (the 'plate'), normally mounted on a post or other support — bearing a traffic sign symbol, whose physical execution is legally prescribed.
Source	https://wetten.overheid.nl/jci1.3:c:BWBR0004825&bijlage=1, https://wetten.overheid.nl/jci1.3:c:BWBR0009104&hoofdstuk=2

Time window

Concept details

Field	Value
Definition	The days and/or hours indicated on a supplementary sign that imply a limitation of the temporal scope of application of the indication on the overlying traffic sign; in particular, for speed limits, the period during which the maximum speed indicated on the traffic sign applies.
Scope	The term 'time window' does not appear as such in the legislation; the legal basis lies in the limitation of the scope and is referred to as a 'time window' by Rijkswaterstaat and CROW. The concept has a dual realization: as a physical supplementary sign beneath a metal sign, or as a time condition within a digital/electronic sign. Not to be confused with 'window times' (loading and unloading times for urban logistics), which are incorporated into the Goods Delivery model.
Source	https://kennisbank.crow.nl/public/gastgebruiker/FVV/Verkeerstekens_(toepassing,_plaatsing_en_uitvoering)/Hoofdstuk_C._Geslotenverklaring/29510, https://wetten.overheid.nl/BWBR0004825/, https://wetten.overheid.nl/BWBR0004826/, https://www.rijkswaterstaat.nl/wegen/wetten-regels-en-vergunningen/verkeerswetten/maximumsnelheid

Direction of travel

Concept details

Field	Value
Definition	The direction in which traffic moves on a carriageway or lane. Relevant for speed limits where a limit applies to only one direction of travel, e.g. at separated carriageways or at signs placed on one side of the road.

Road

Concept details

Field	Value
Definition	to be decided

Line geometry

Concept details

Field	Value
Definition	A one-dimensional geometric primitive representing a line: a continuous curve, measurable in length within a coordinate system, composed of one or more connected line segments in which the end point of each segment — except the last — coincides with the start point of the next. Line geometry is used to spatially represent linear features such as a road; designated GM_Curve in ISO 19107 and Curve/LineString in OGC Simple Features
Source	http://www.opengis.net/ont/sf, https://docs.geostandaarden.nl/nen3610/gimeg/, https://modellen.geostandaarden.nl/def/nen3610-2022/

Area geometry

Concept details

Field	Value
Definition	to be decided

Traffic decision

Concept details

Field	Value
Definition	Written decision by the competent authority deciding on the placement, modification, or removal of traffic signs and supplementary signs designated by Order in Council insofar as this creates or modifies a command or prohibition, or on measures on or along the road that lead to a restriction or expansion of the number of categories of road users.
Scope	Publication takes place pursuant to Article 26 of the BABW via the KOOP Traffic Decisions application in the Government Gazette (National Government) or the local publication journal (municipality, province, water board) since the entry into force of the Electronic Publications Act on 1 July 2021. Requirements for justification follow from Article 21 of the BABW; consultation with the Chief of Police follows from Article 24 of the BABW.
Source	https://wetten.overheid.nl/jci1.3:c:BWBR0006622&artikel=12, https://wetten.overheid.nl/jci1.3:c:BWBR0006622&artikel=15, https://wetten.overheid.nl/jci1.3:c:BWBR0006622&artikel=18, https://wetten.overheid.nl/jci1.3:c:BWBR0006622&artikel=1:3, https://wetten.overheid.nl/jci1.3:c:BWBR0006622&artikel=21, https://wetten.overheid.nl/jci1.3:c:BWBR0006622&artikel=26

N-road

Concept details

Field	Value
Definition	The area outside the built-up area established by the municipal council. Outside the built-up area the maximum speeds are 130 km/h on motorways, 100 km/h on expressways and 80 km/h on other roads, unless otherwise indicated by traffic signs.

Relations per concept

Outgoing semantic relations only — these stay inside the model.

Outer regime

Semantic relations

Type	Target label	Target concept
broader	Geometry	https://ndwnu.github.io/its-gegevenscatalogus/id/9a2ca40b-96c1-4526-a5c5-fda1a468fc5b/concept-1771237797860
narrower	Inner regime	https://ndwnu.github.io/its-gegevenscatalogus/id/9a2ca40b-96c1-4526-a5c5-fda1a468fc5b/concept-1771237259923

Residential Area

Semantic relations

Type	Target label	Target concept
broader	Area geometry	https://ndwnu.github.io/its-gegevenscatalogus/id/9a2ca40b-96c1-4526-a5c5-fda1a468fc5b/concept-1772186704395

Speedlimit

Semantic relations

Type	Target label	Target concept
related	Geometry	https://ndwnu.github.io/its-gegevenscatalogus/id/9a2ca40b-96c1-4526-a5c5-fda1a468fc5b/concept-1771237797860
related	Indication	https://ndwnu.github.io/its-gegevenscatalogus/id/9a2ca40b-96c1-4526-a5c5-fda1a468fc5b/concept-1771236172013

Expressway

Semantic relations

Type	Target label	Target concept
broader	Line geometry	https://ndwnu.github.io/its-gegevenscatalogus/id/9a2ca40b-96c1-4526-a5c5-fda1a468fc5b/concept-1772186695677

Motorway

Semantic relations

Type	Target label	Target concept
broader	Line geometry	https://ndwnu.github.io/its-gegevenscatalogus/id/9a2ca40b-96c1-4526-a5c5-fda1a468fc5b/concept-1772186695677

Carriageway

Semantic relations

Type	Target label	Target concept
related	Direction of travel	https://ndwnu.github.io/its-gegevenscatalogus/id/9a2ca40b-96c1-4526-a5c5-fda1a468fc5b/concept-1772186214125
related	Lane	https://ndwnu.github.io/its-gegevenscatalogus/id/9a2ca40b-96c1-4526-a5c5-fda1a468fc5b/concept-1771235638908

Lane

Semantic relations

Type	Target label	Target concept
related	Road section	https://ndwnu.github.io/its-gegevenscatalogus/id/9a2ca40b-96c1-4526-a5c5-fda1a468fc5b/concept-1771235085917

Type	Target label	Target concept
related	Direction of travel	https://ndwnu.github.io/its-gegevenscatalogus/id/9a2ca40b-96c1-4526-a5c5-fda1a468fc5b/concept-1772186214125

Vehicle category

Semantic relations

Type	Target label	Target concept
related	Speedlimit	https://ndwnu.github.io/its-gegevenscatalogus/id/9a2ca40b-96c1-4526-a5c5-fda1a468fc5b/concept-1771235237134

Start Indication

Semantic relations

Type	Target label	Target concept
broader	Indication	https://ndwnu.github.io/its-gegevenscatalogus/id/9a2ca40b-96c1-4526-a5c5-fda1a468fc5b/concept-1771236172013

End Indication

Semantic relations

Type	Target label	Target concept
broader	Indication	https://ndwnu.github.io/its-gegevenscatalogus/id/9a2ca40b-96c1-4526-a5c5-fda1a468fc5b/concept-1771236172013

Indication

Semantic relations

Type	Target label	Target concept
narrower	Actuator	https://ndwnu.github.io/its-gegevenscatalogus/id/9a2ca40b-96c1-4526-a5c5-fda1a468fc5b/concept-1771236355335
narrower	Traffic sign	https://ndwnu.github.io/its-gegevenscatalogus/id/9a2ca40b-96c1-4526-a5c5-fda1a468fc5b/concept-1772185159176
related	Inner regime	https://ndwnu.github.io/its-gegevenscatalogus/id/9a2ca40b-96c1-4526-a5c5-fda1a468fc5b/concept-1771237259923
related	Supplementary Sign	https://ndwnu.github.io/its-gegevenscatalogus/id/9a2ca40b-96c1-4526-a5c5-fda1a468fc5b/concept-1771236462476

Condition

Semantic relations

Type	Target label	Target concept
related	Time window	https://ndwnu.github.io/its-gegevenscatalogus/id/9a2ca40b-96c1-4526-a5c5-fda1a468fc5b/concept-1772186160984
related	Weather Conditions	https://ndwnu.github.io/its-gegevenscatalogus/id/9a2ca40b-96c1-4526-a5c5-fda1a468fc5b/concept-1771236028582

Supplementary Sign

Semantic relations

Type	Target label	Target concept
related	Condition	https://ndwnu.github.io/its-gegevenscatalogus/id/9a2ca40b-96c1-4526-a5c5-fda1a468fc5b/concept-1771236446358

Built-Up Area

Semantic relations

Type	Target label	Target concept
broader	Area geometry	https://ndwnu.github.io/its-gegevenscatalogus/id/9a2ca40b-96c1-4526-a5c5-fda1a468fc5b/concept-1772186704395

Inner regime

Semantic relations

Type	Target label	Target concept
broader	Geometry	https://ndwnu.github.io/its-gegevenscatalogus/id/9a2ca40b-96c1-4526-a5c5-fda1a468fc5b/concept-1771237797860

Geometry

Semantic relations

Type	Target label	Target concept
related	Start Indication	https://ndwnu.github.io/its-gegevenscatalogus/id/9a2ca40b-96c1-4526-a5c5-fda1a468fc5b/concept-1771236090614
related	End Indication	https://ndwnu.github.io/its-gegevenscatalogus/id/9a2ca40b-96c1-4526-a5c5-fda1a468fc5b/concept-1771236142216

Speed Limit Data Type

Semantic relations

Type	Target label	Target concept
related	Speedlimit	https://ndwnu.github.io/its-gegevenscatalogus/id/9a2ca40b-96c1-4526-a5c5-fda1a468fc5b/concept-1771235237134

Traffic sign

Semantic relations

Type	Target label	Target concept
related	Actuator	https://ndwnu.github.io/its-gegevenscatalogus/id/9a2ca40b-96c1-4526-a5c5-fda1a468fc5b/concept-1771236355335

Road

Semantic relations

Type	Target label	Target concept
narrower	Expressway	https://ndwnu.github.io/its-gegevenscatalogus/id/9a2ca40b-96c1-4526-a5c5-fda1a468fc5b/concept-1771235313383
narrower	Motorway	https://ndwnu.github.io/its-gegevenscatalogus/id/9a2ca40b-96c1-4526-a5c5-fda1a468fc5b/concept-1771235331951
narrower	N-road	https://ndwnu.github.io/its-gegevenscatalogus/id/9a2ca40b-96c1-4526-a5c5-fda1a468fc5b/concept-1772187859423
related	Carriageway	https://ndwnu.github.io/its-gegevenscatalogus/id/9a2ca40b-96c1-4526-a5c5-fda1a468fc5b/concept-1771235620358

Line geometry

Semantic relations

Type	Target label	Target concept
broader	Geometry	https://ndwnu.github.io/its-gegevenscatalogus/id/9a2ca40b-96c1-4526-a5c5-fda1a468fc5b/concept-1771237797860

Area geometry

Semantic relations

Type	Target label	Target concept
broader	Geometry	https://ndwnu.github.io/its-gegevenscatalogus/id/9a2ca40b-96c1-4526-a5c5-fda1a468fc5b/concept-1771237797860

Traffic decision

Semantic relations

Type	Target label	Target concept
related	Speedlimit	https://ndwnu.github.io/its-gegevenscatalogus/id/9a2ca40b-96c1-4526-a5c5-fda1a468fc5b/concept-1771235237134

N-road

Semantic relations

Type	Target label	Target concept
broader	Line geometry	https://ndwnu.github.io/its-gegevenscatalogus/id/9a2ca40b-96c1-4526-a5c5-fda1a468fc5b/concept-1772186695677